Towards Doped Cryogenic Rare Gas Crystals for Precision Measurement

Isaac Cross[1], Daniel Lutes[1], Joel Adhiambo[1], Brian Odom[1]

1. Department of Physics and Astronomy, Northwestern University, Evanston, IL, USA

Center for **Fundamental Physics**

Argonne NATIONAL LABORATORY

Motivation

Atomic systems in beams and traps have made massive leaps in the past few decades as clocks, quantum sensors, or for precision BSM physics searches. Scalability is a major issue for applications across the board. **Doped noble gas crystals can address this for some applications by hosting up to 10^{19} dopants/mm cubed. Low polarizability** enables longer coherence times than other solid-state systems, **and transparency enables standard AMO state preparation and readout techniques**. We will explore channels of spin relaxation by addressing dopants optically while studying crystal structure with x-rays in parallel to deduce the connection between dopant line width and crystal structure. While also exploring the growth parameter space of these crystals and their influence on structure and linewidth. In this poster we present both our current work towards a tunable dual-species Rb-Ar beam source and crystal growth schemes and success.

Comparison to Existing Platforms

Existing Platforms

Beams:



T2* = 2-3 ms
N = 10^{12}

ACME Collaboration
Nature **562**, 355-360 (2018)

Traps:

JILA

T2* = 5 s
N = 10^{7}

JILA. Eric Cornell
slides ITAMP 2024

General EDM sensitivity: $\sigma \propto \frac{1}{T_2^* \sqrt{N}}$

Doped Noble Gas Crystals

Cs in He:
T2* = 0.1 s
10^9 atoms/cm^3

Kanorsky et al.,
Phys. Rev. A **54**, R1010 (1996)

Demonstrates one can achieve long coherence times in a cryogenic crystal →

Rb in Ne:
T2 = 0.1 s
T2* = 10 μs
10^{16} atoms/cm^3

J.D. Weinstein et al.,
Phys. Rev. A **104**, 032611 (2021)

Not enough dopants due to high helium pressure required for growth →

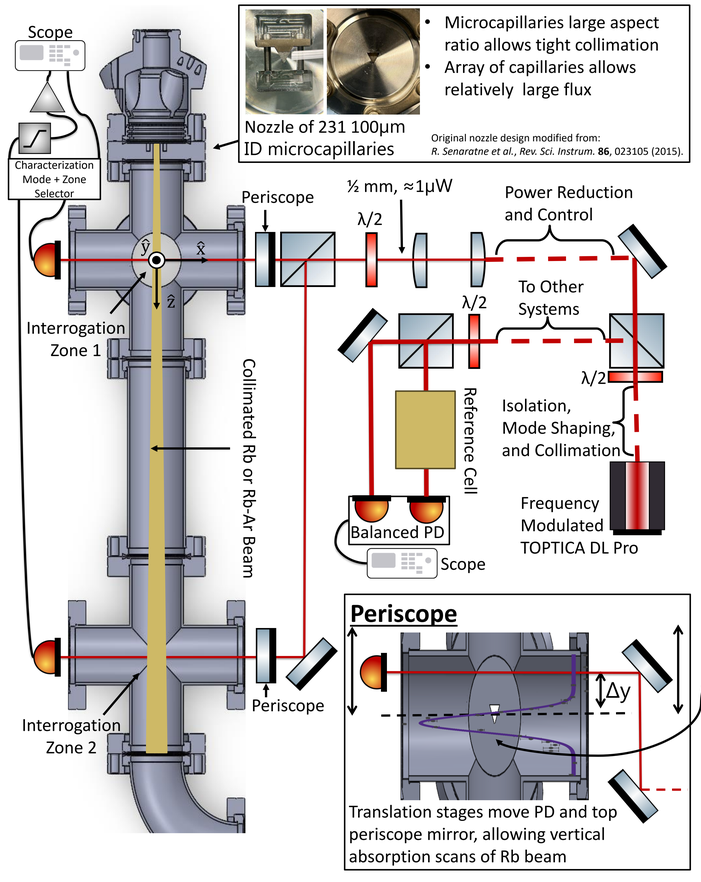
→ Many dopants
↓
Large discrepancy between T2 and T2* due to inhomogeneous broadening, thought to be from grain boundaries
↓
Will address this by growing single crystal seeds and then performing epitaxy

Beam Source

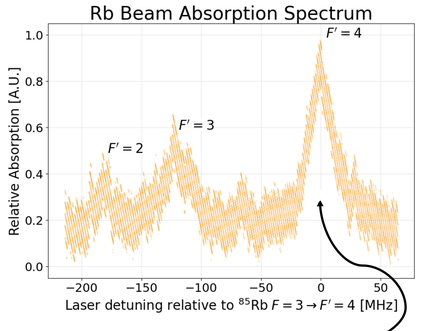
Overview

Our beam source is designed to transport a collimated dual-species beam of Rb and Ar to our growth substrate. Our beam sets key parameters such as dopant density and growth rate. Characterization of a Rb beam has been done shown in this section we are currently introducing Ar.

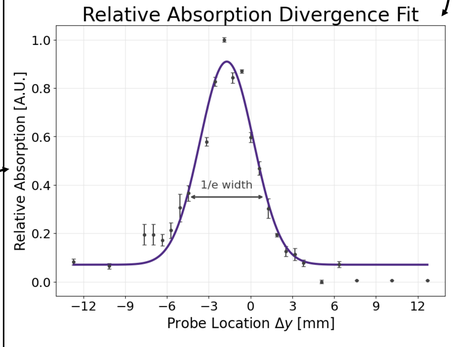
Beam Characterization Setup



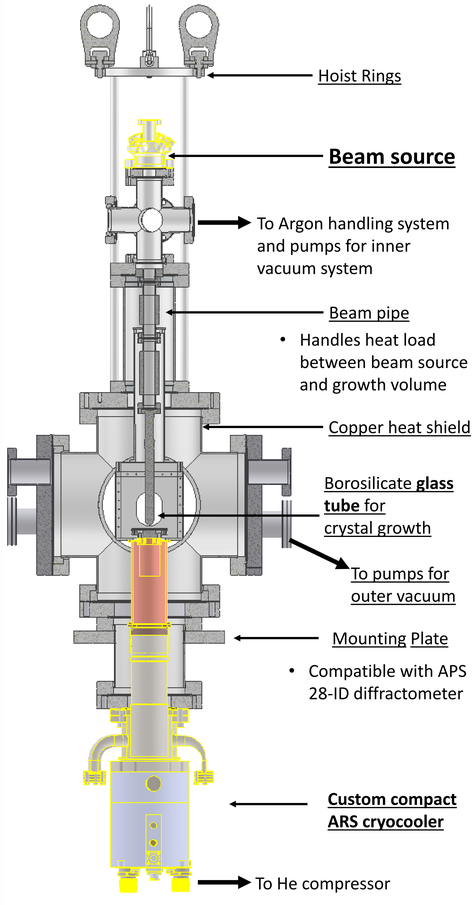
Characterization Procedure



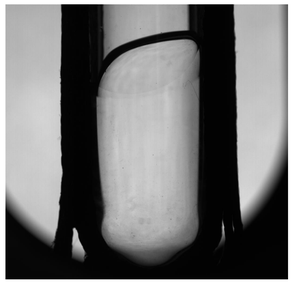
- Scan over ^{85}Rb $F = 3 \rightarrow F' = 4$
- Position probe beam and PD at variable Δy (Look Periscope) and take absorption data
- Fit with normalized Lorentzian to extract peak height
- Fit relative absorption vs Δy with gaussian to extract 1/e width for zone 1 and 2
- half angle divergence: $(\alpha)_{1/e} = \mathbf{0.48 \pm 0.1}°$
- Capture efficiency: $\mathbf{68 \pm 7}\%$
- Flux: $\mathbf{1.78 \times 10^{10}}$ **atoms/s**



Crystal Growth



Pictures of Cm scale transparent argon crystals



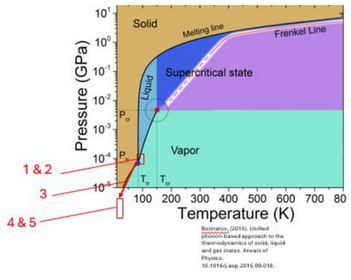
Left: thorcam



Right: iphone photo

Overview

- Optically transparent cm scale argon crystals have been grown using the Bridgeman-Stockbarger method at cryogenic temperatures
- Portable cryostat supports optical access and is compatible with APS 28ID Beamline
- Beam source and cryostat Integration is on the horizon
- Will use to study inhomogeneous broadening in dopant ensemble



Bridgman-Stockbarger

Ar liquid → "Pulling from melt" → 1. Condense → 2. Freeze → 3. Cool

Vapor Deposition

Rb & Ar beam → 4. Dope → Optical T2* + extras → X-rays → 5. Measure

References

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- U. Dargyte, D. M. Lancaster, and J. D. Weinstein, *Optical and spin-coherence properties of rubidium atoms trapped in solid neon*, Phys. Rev. A **104**, 032611 (2021).

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